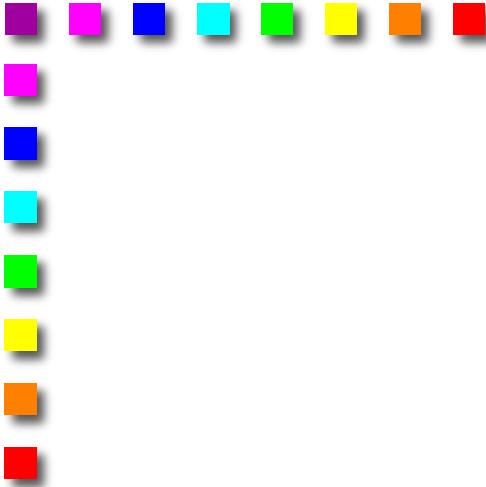


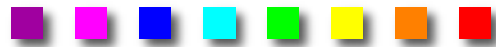


VPN

Virtual Private Network

Mario Baldi
Politecnico di Torino





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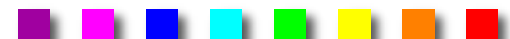
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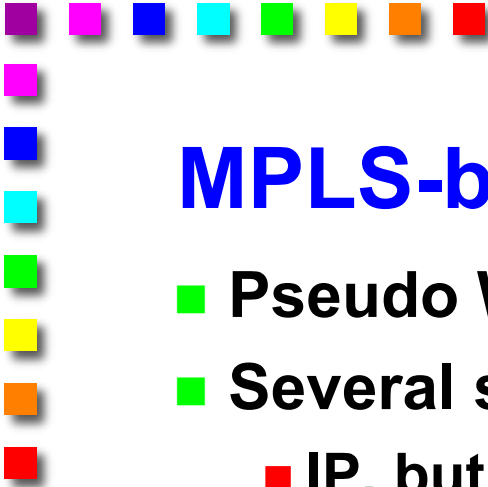
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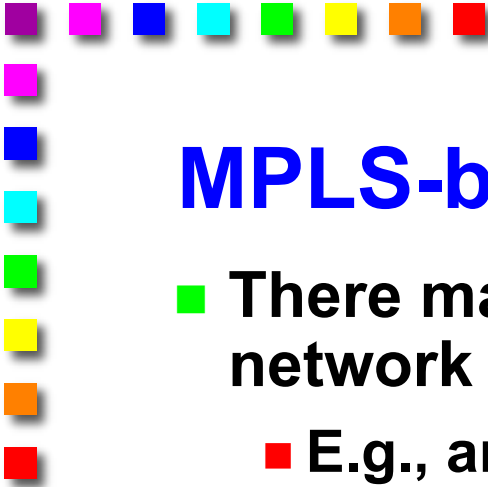
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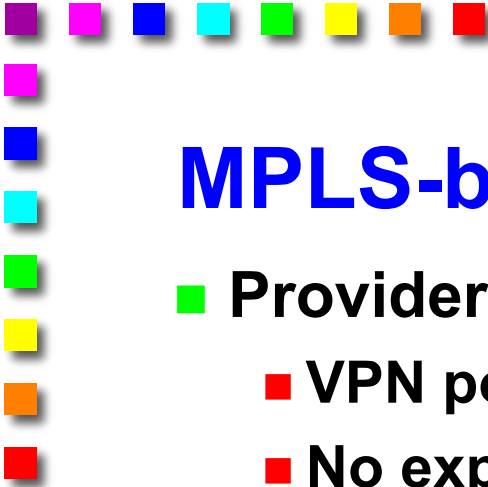
MPLS-based Layer 2 VPNs: PWE3

- Pseudo Wire Emulation End-to-End
 - Several services on the same network:
 - IP, but also leased lines, frame relay, ATM, Ethernet
 - Customer edge (CE) device features native service interface
 - Traffic is carried through an LSP between CEs
 - Two labels
 - External – for routing within the network
 - Identifies access point to the network
 - Internal - multiplexing of several users/services at the same access point
- 



MPLS-based Layer 2 VPNs: PWE3

- There may be aggregation devices inside the network
 - E.g., an ATM switch inside the service provider network switching traffic between users
 - LSP ended on the device
- Mainly manual LSP setup
- Proposals exist for deployment of LDP and BGP



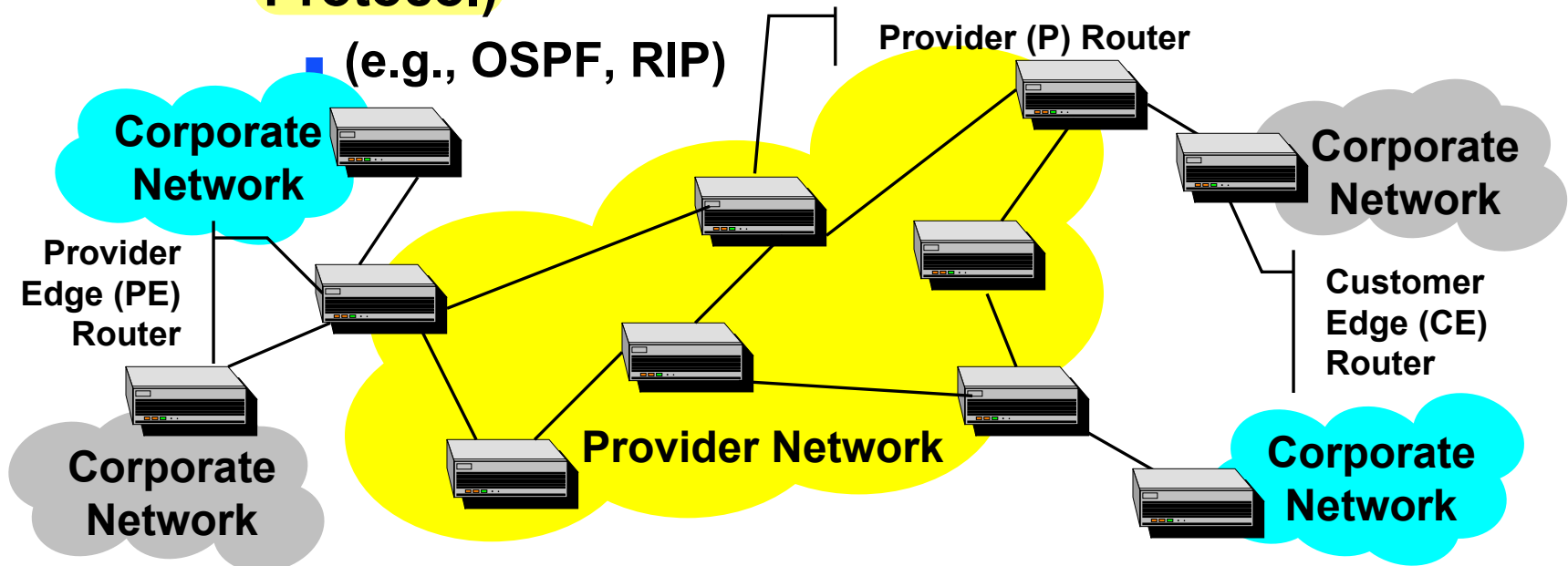
MPLS-based Layer 3 VPNs

- **Provider provisioned solutions**
 - VPN policies implemented by Service Provider
 - No experience needed on the Customer side
 - **Scalability**
 - Large scale deployments
 - **Two alternative solutions**
 - **RFC2547bis (BGP)**
 - Initially supported by Cisco Systems
 - Currently most widely deployed approach
 - **Virtual router**
 - Initially supported by Nortel and Lucent
- 

MPLS VPN Components

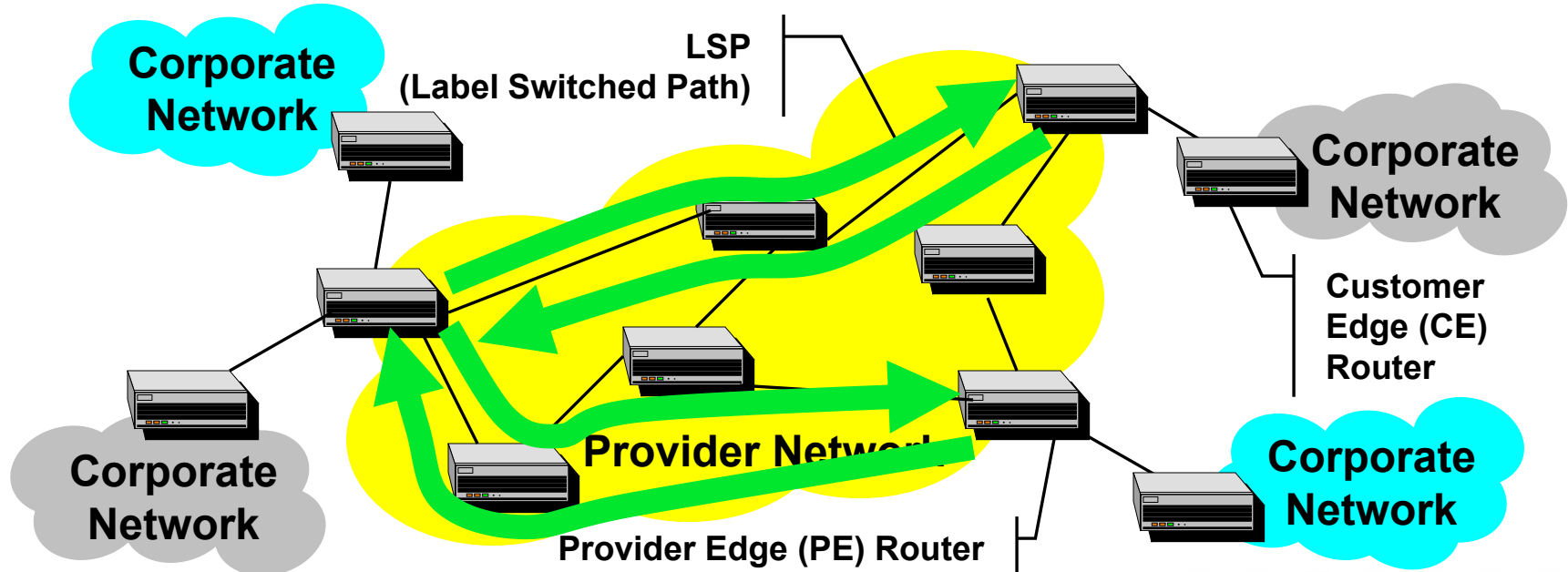
- **CE router creates adjacency with PE router**
 - It advertises its destinations
 - It receives advertisements of other VPN destinations
 - Static routing, or IGP (Interior Gateway Protocol)

■ (e.g., OSPF, RIP)



MPLS VPN Components

- P routers have routes to PE routers only
- PEs setup LSPs among themselves
 - LDP and/or RSVP (and/or I-BGP)
 - E.g., topology-based label binding



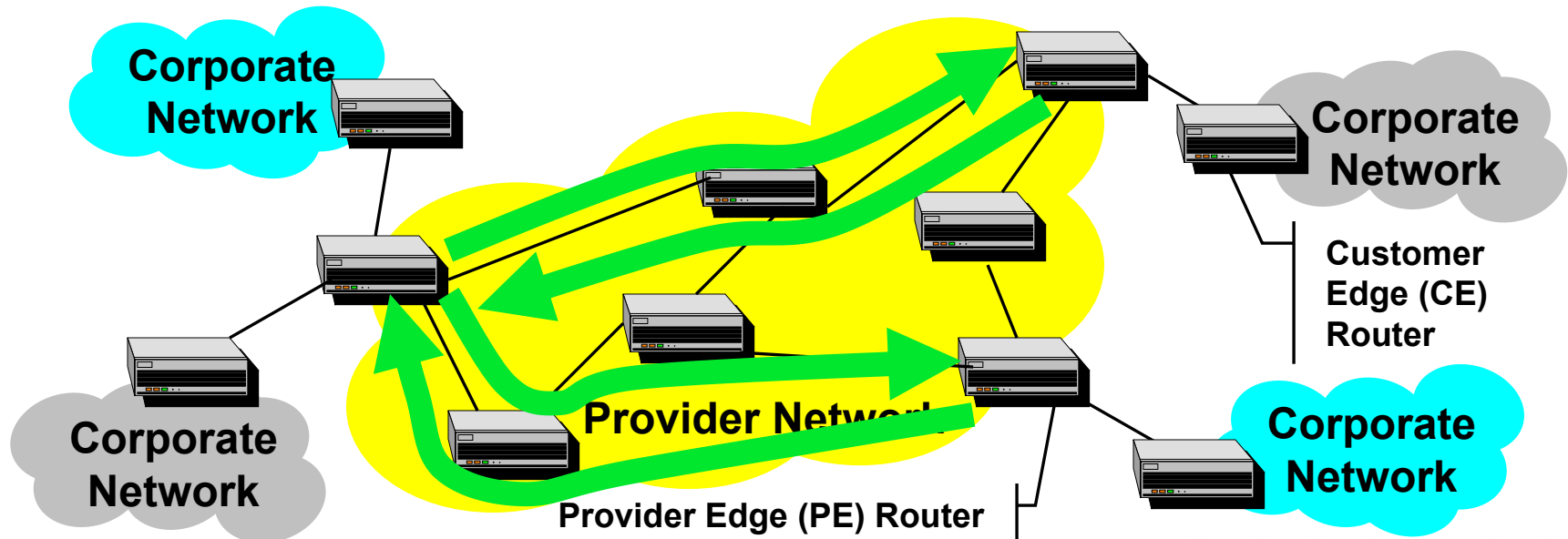
MPLS VPN Components

■ PE routers

■ Exchange routing information

- I-BGP (Interior-Border Gateway Protocol) in BGP-based solution
- IGP in VR solution

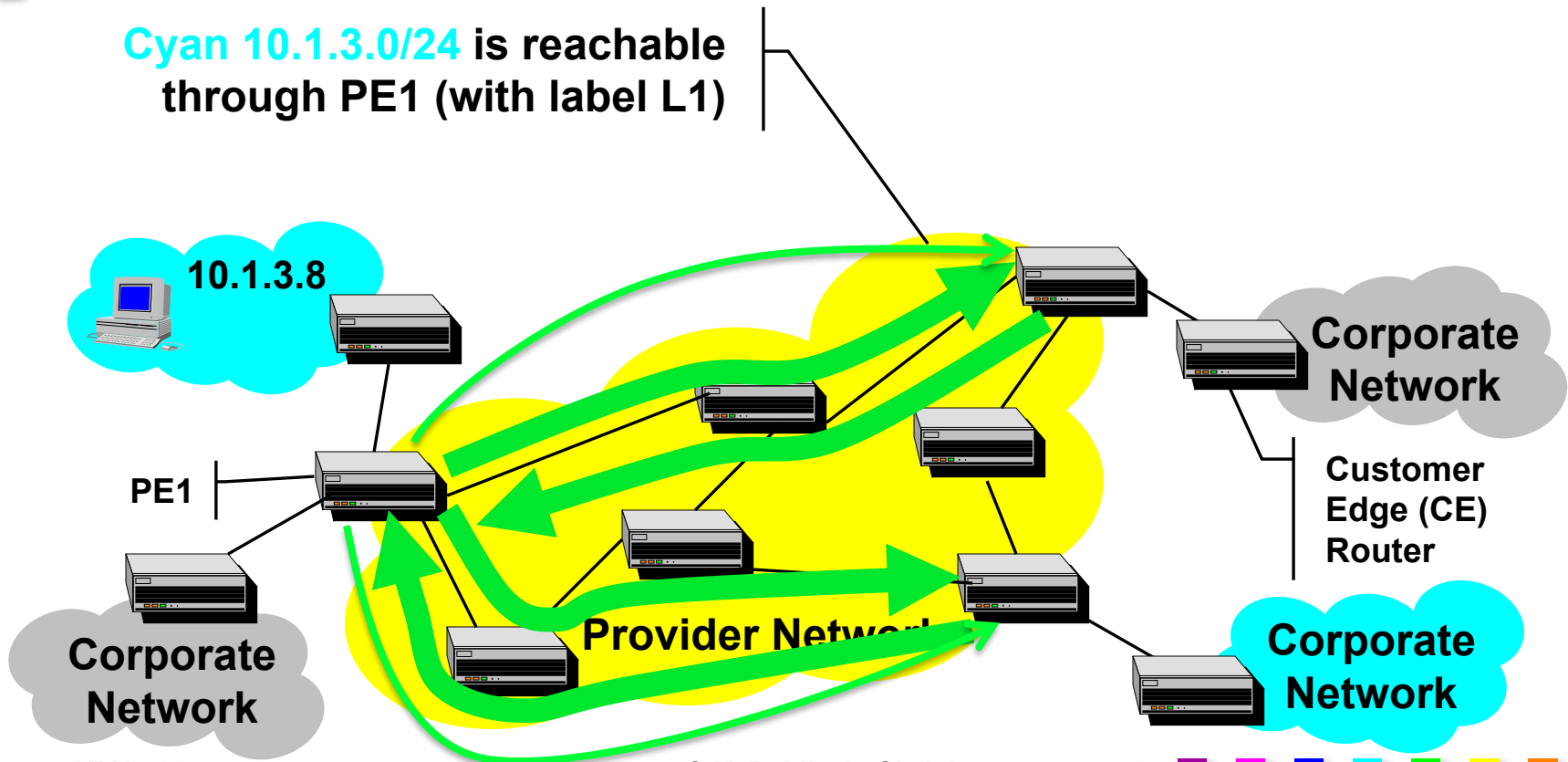
■ PE keeps routes only for VPNs connected to it



MPLS/BGP VPN Components

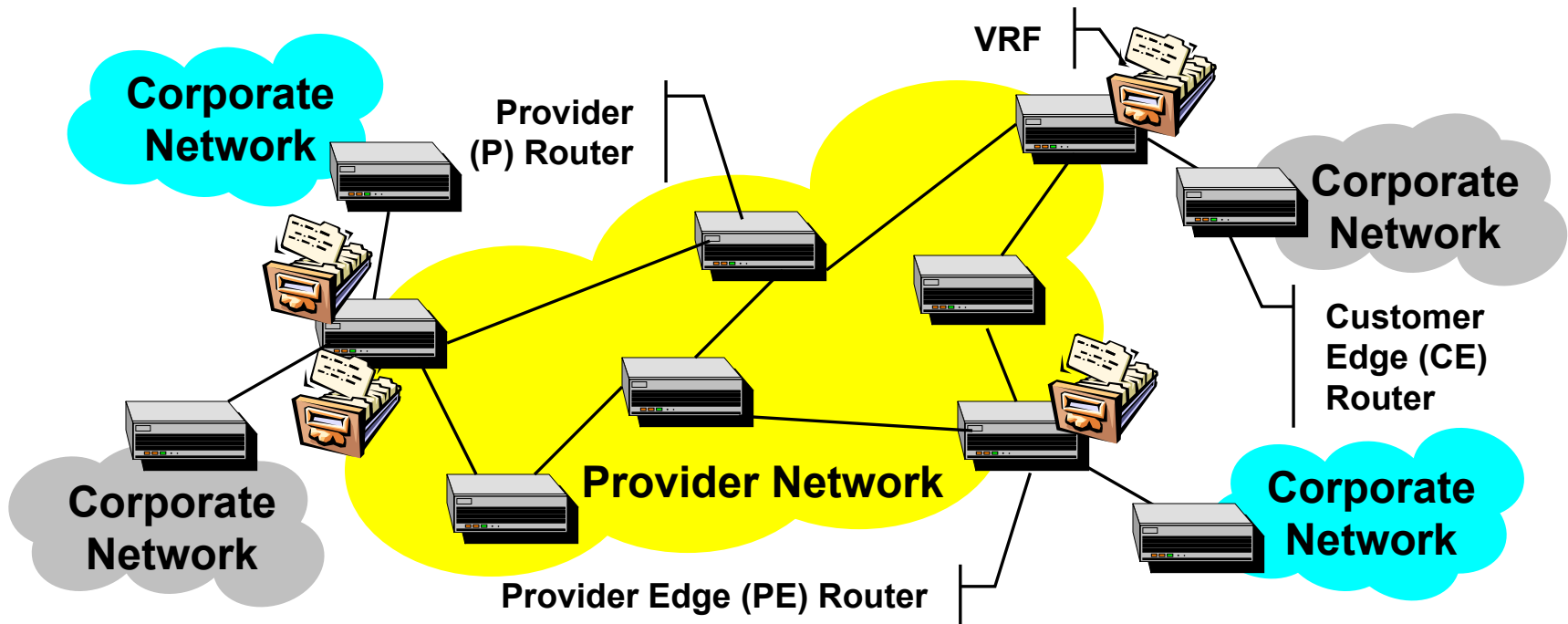
■ PE routing exchanges (with I-BGP)

Cyan 10.1.3.0/24 is reachable
through PE1 (with label L1)



MPLS/BGP VPN Components

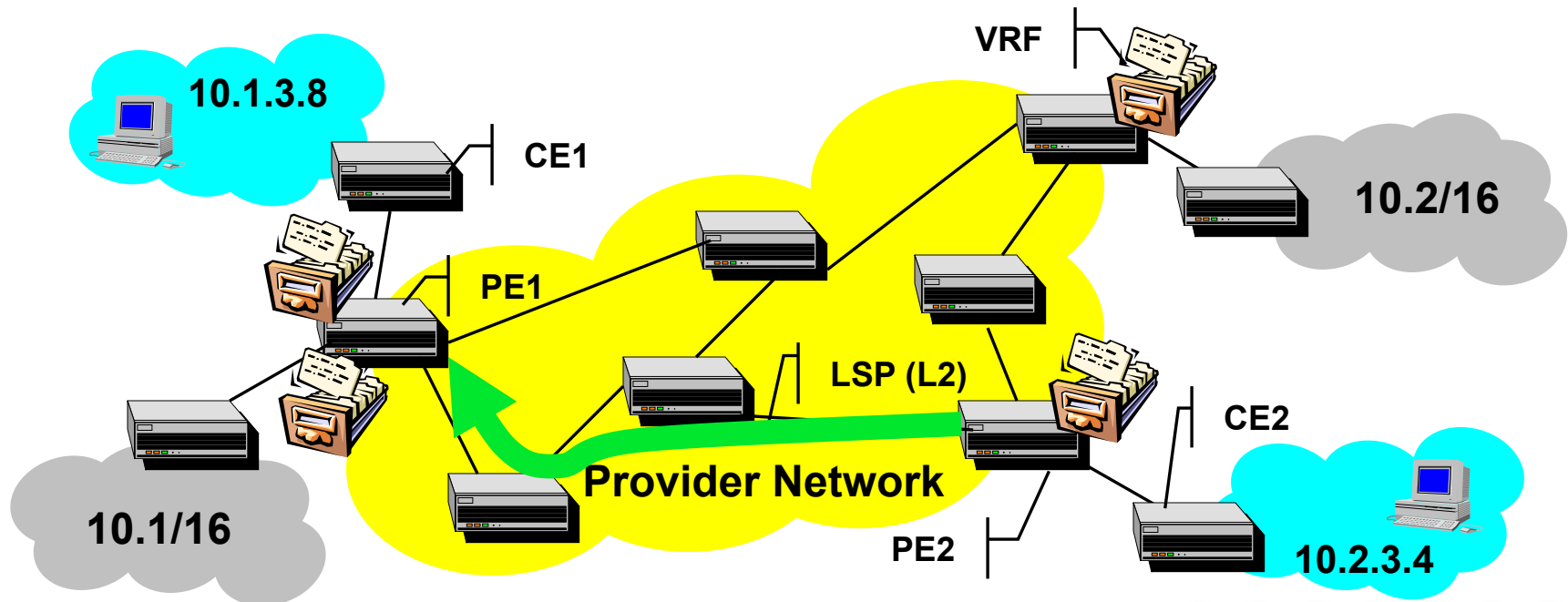
- **VRF (VPN Routing and Forwarding) table**
 - Associated to one or more (non-MPLS) ports
 - Forwarding information to be used for traffic received through the port



Packet Routing

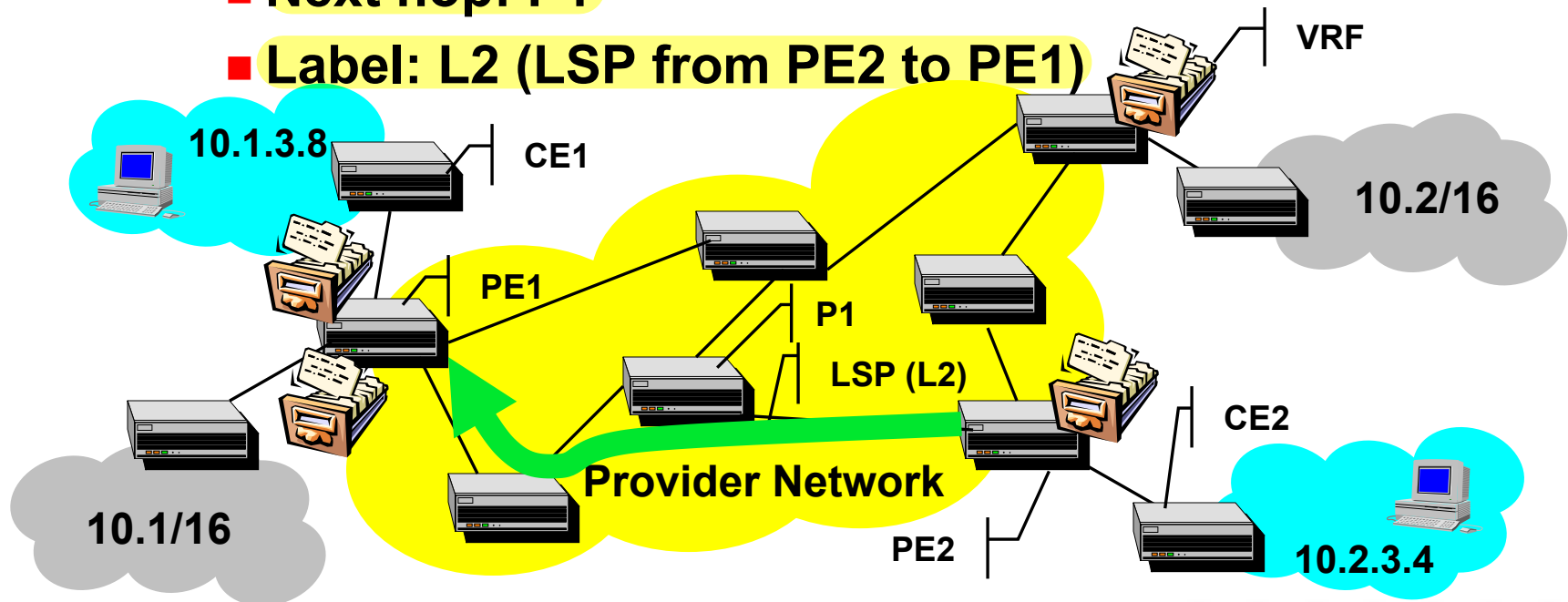
Packet from cyan 10.2.3.4 to 10.1.3.8

- Default gateway → PE2 router



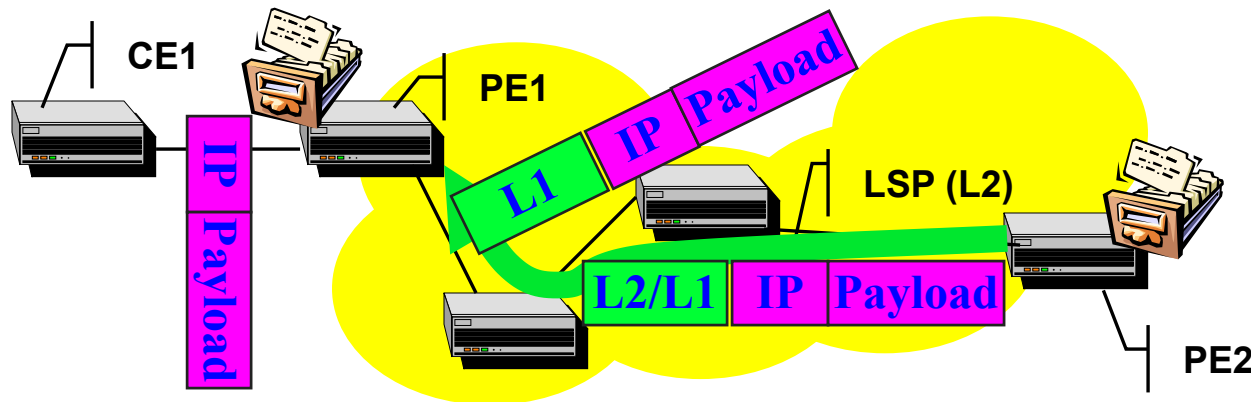
Packet Routing

- PE2 looks-up 10.1.3.8 in cyan VRF
 - Next hop: PE1
 - Label: L1 (distributed by PE1 for cyan 10.1.3.0/24)
- PE2 looks up PE1 in main table
 - Next hop: P1
 - Label: L2 (LSP from PE2 to PE1)



Packet Routing

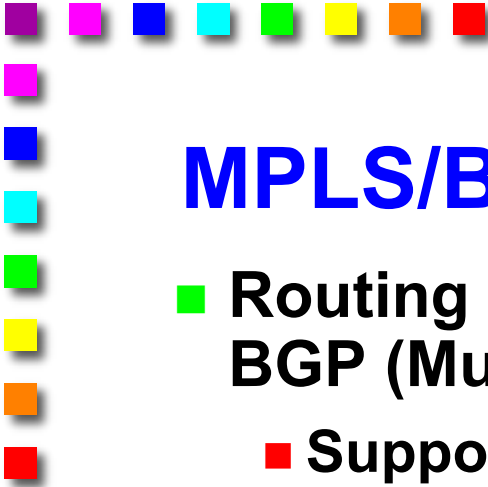
- PE2 has pushed L1 and L2 on label stack
- P routers forward packet to PE1 using L2
- Last hop before PE1 pops L2 (PHP)
- PE1 receives packet with L1
 - PE1 pops L1: plain IP packet
 - PE1 uses L1 to route packet to proper output interface





Benefits

- No constraints on addressing plan
 - Address uniqueness only within VPN
 - CE routers do not exchange information with each other
 - Customer does not manage backbone
 - Providers do not have one virtual backbone per customer
 - VPN can span multiple providers
 - Security equivalent to Frame relay or ATM
 - Traffic isolation
 - No cryptography (confidentiality)
 - QoS supported through experimental bits in MPLS header
- 



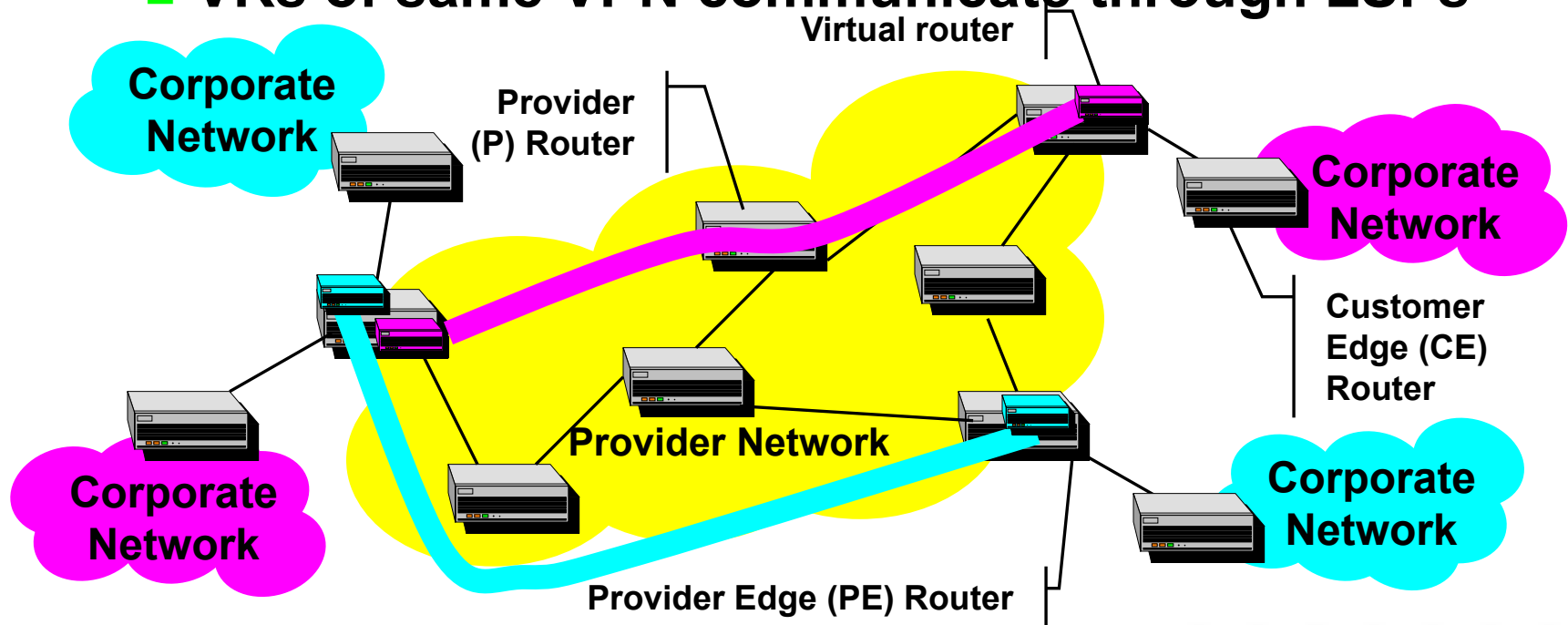
MPLS/BGP VPNs

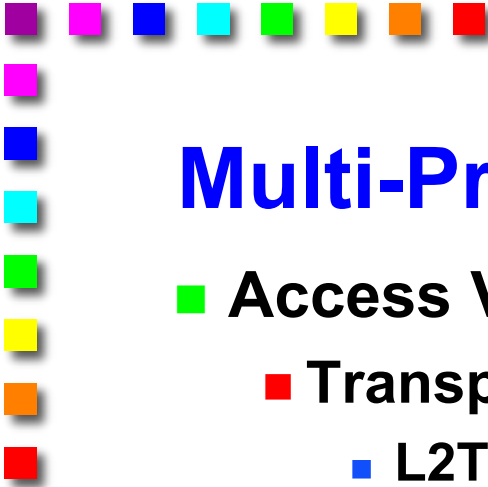
- Routing exchange at edges based on MP-BGP (Multi-protocol BGP)
 - Support for addresses of different families
- Route filtering
 - PE routers determine which routes to install in VRF
- Support for overlapping address spaces
 - VPN-IPv4 Address family
 - Route Distinguisher + IPv4 address



MPLS/Virtual Router VPNs

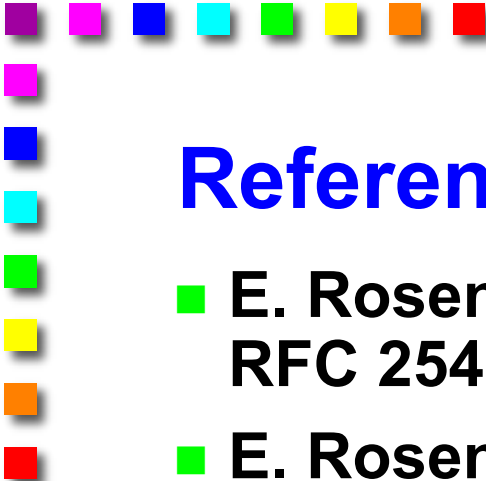
- PEs execute a (virtual) router instance for each VPN
- Each VR instance has separate data structures
- VRs of same VPN communicate through LSPs





Multi-Protocol Support

- Access VPN
 - Transparent
 - L2TP and PPTP
 - Overlay (IPsec based)
 - Generic Routing Encapsulation (GRE)
 - Transport any layer 3 protocol within IP
 - Peer (MPLS based)
 - Built in MPLS (*Multi-Protocol* Label Switching)
- 



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